THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The net of a cone consists of: 1

(a) one circle and one triangle **(b)** one square and four triangles
(c) two circles and a rectangle **(d)** one circle and one sector

2. The front view of a cylinder standing on its base is a: 1

(a) circle **(b)** triangle **(c)** square always **(d)** rectangle

3. A polyhedron has 6 vertices and 9 edges. The number of faces is: 1

(a) 5 **(b)** 17 **(c)** 3 **(d)** 13

4. A Koch curve starts with one segment. The number of segments at stage 3 is: 1

(a) 12 **(b)** 27 **(c)** 64 **(d)** 63

5. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 4 is: 1

(a) $\frac{1}{256}$ **(b)** $\frac{80}{256}$ **(c)** $\frac{12}{16}$ **(d)** $\frac{81}{256}$

6. A cube of side 5 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 36 **(b)** 27 **(c)** 8 **(d)** 54

7. The net of a triangular prism consists of: 1

(a) five rectangles **(b)** two triangles and three rectangles
(c) three triangles and two rectangles **(d)** four triangles and one rectangle

8. The front view of a cone standing on its base is a: 1

(a) rectangle **(b)** triangle **(c)** circle **(d)** pentagon

9. A polyhedron has 7 vertices and 12 edges. The number of faces is: 1

(a) 17 **(b)** 7 **(c)** 21 **(d)** 5

10. A Koch curve starts with one segment. The number of segments at stage 4 is: 1

(a) 256 **(b)** 255 **(c)** 81 **(d)** 16

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Verify Euler's formula for a decagonal pyramid. 2

12. A solid has 10 faces and 24 edges. Use Euler's formula to find its number of vertices. 2

13. A prism has 33 edges. Find the number of sides of its base and the number of faces. 2

14. Verify Euler's formula for a dodecagonal pyramid. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 5 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3
(i) exactly two faces painted
(ii) exactly one face painted.

16. A Koch curve starts with a straight line of length 9 cm. How many segments are there at stage 2, how long is each, and what is the total length? 3

17. A Sierpinski triangle starts with a triangle of area 256 cm². Find the shaded area after stage 3 and the area removed. 3

OR

A cube of side 6 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3
(i) exactly two faces painted
(ii) exactly one face painted.

18. A Koch curve starts with a straight line of length 18 cm. How many segments are there at stage 2, how long is each, and what is the total length? 3

OR

A Sierpinski triangle starts with a triangle of area 512 cm². Find the shaded area after stage 3 and the area removed. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the number of faces, vertices and edges of 5
(i) a pentagonal prism and
(ii) a pentagonal pyramid, and verify Euler's formula for both.

OR

A cube of side 12 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 1728. 5

20. Find the number of faces, vertices and edges of 5
(i) a hexagonal prism and
(ii) a hexagonal pyramid, and verify Euler's formula for both.

OR

A cube of side 15 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 3375. 5

END OF QUESTION PAPER Total: 40 Marks